

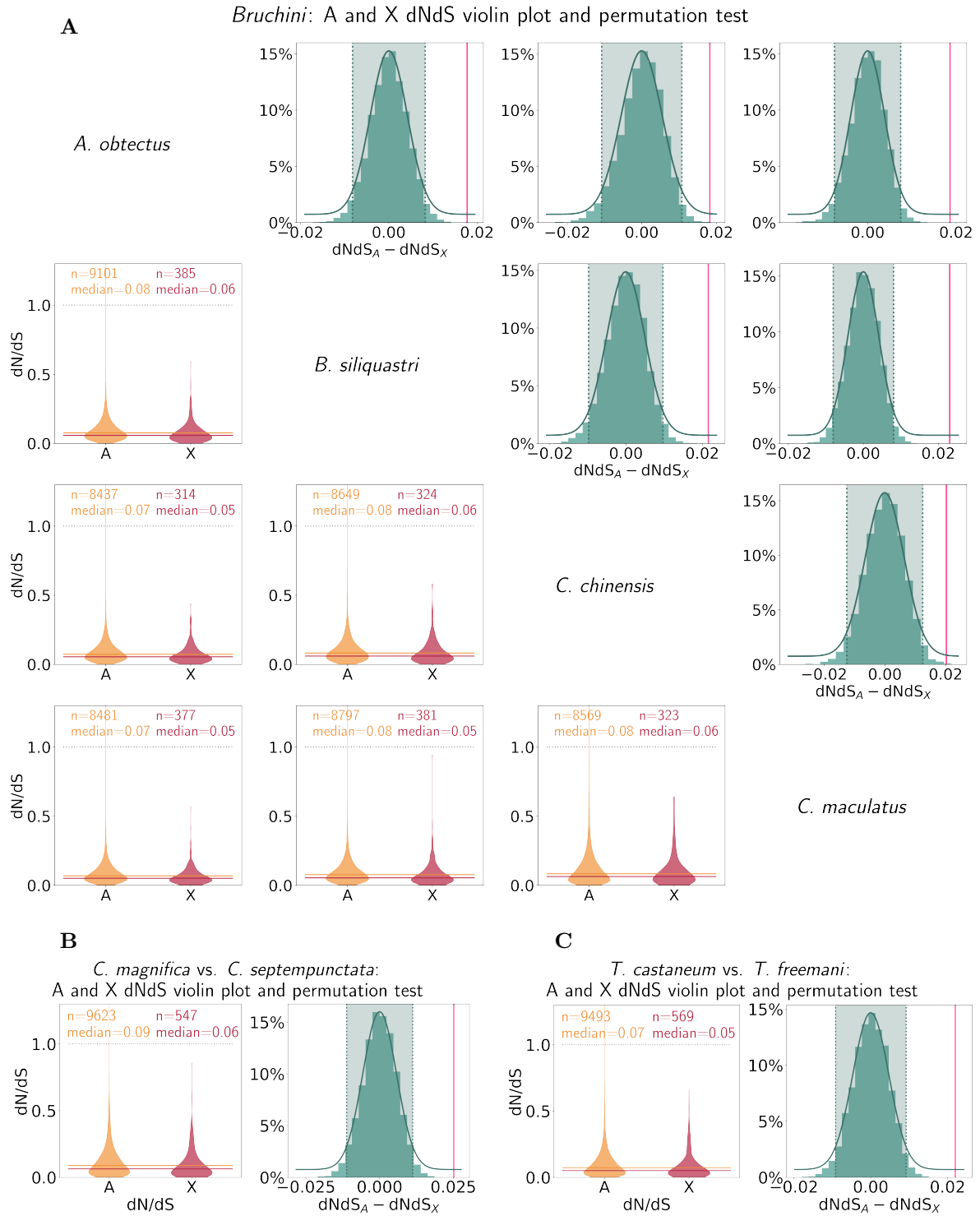


Figure 1: dN/dS of autosomal (A) and X-linked (X) genes Pairwise comparisons between members of three species lineages with violin plots of dN/dS values for X-linked and autosomal 1-to-1 orthologs. Permutation tests ($n=10000$, visualized with histograms, pink line shows measured median difference) show that dN/dS is significantly lower for X-linked orthologs in all comparisons within *Bruchini* (A), *Coccinella* (B), and *Tribolium* (C).

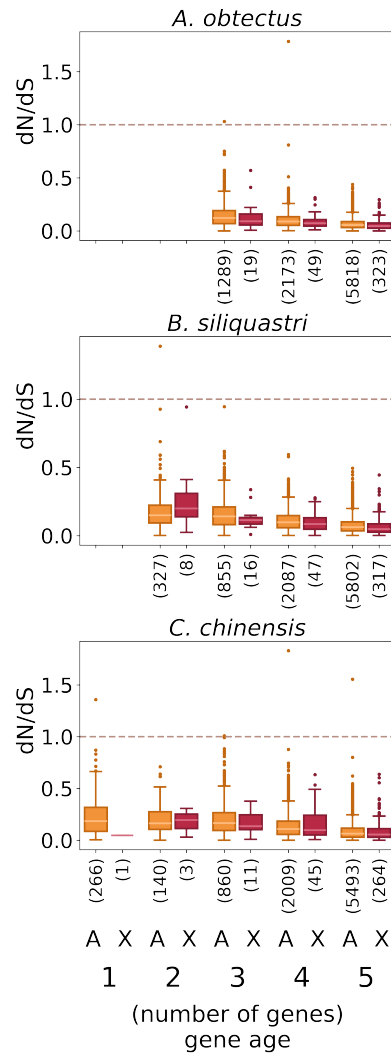


Figure 2: **Relative gene age:** Pairwise comparisons of *C. maculatus* with *A. obtectus*, *B. siliquastris* and *C. chinensis* respectively. The X-axis shows sample size in parentheses, as well as phylogenetic gene age (less conserved orthologs of age 1 to most conserved orthologs of age 5), and autosomal or X-linkage of orthologs (A or X). The box plots show dN/dS for autosomal (yellow) or X-linked (red) orthologs by gene age for *C. maculatus* compared to three other *Bruchini* species.

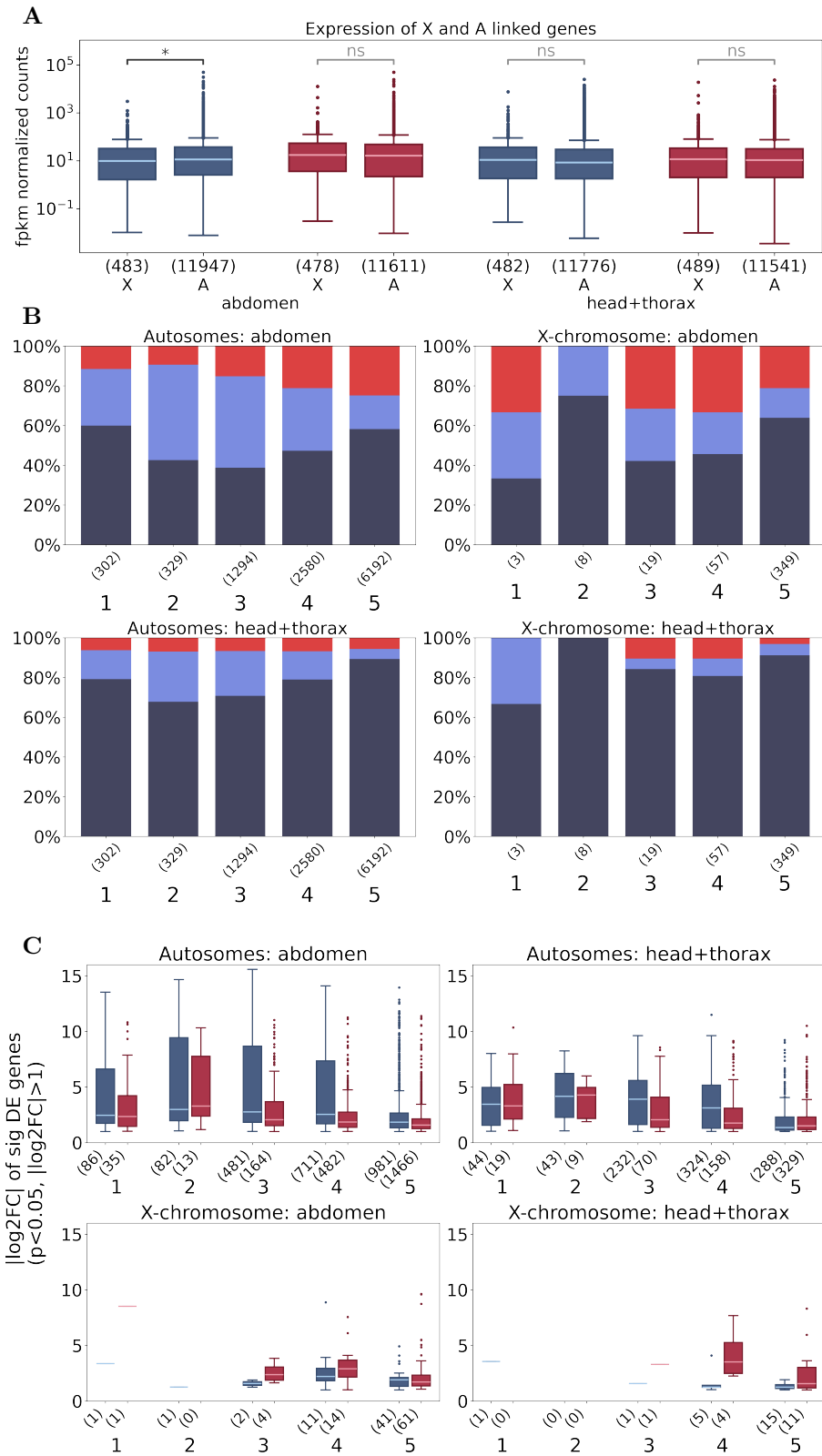


Figure 3: (**A**): **Median expression** (FPKM-normalized mean > 0 across three biological replicates) in male (blue) and female (red) samples for somatic (abdomen) and reproductive (head + thorax) tissues. Mann-Whitney-U test for significant differences is applied and significant differences are marked with an asterisk. (**B,C**): **Gene age by sex-bias analysis**. (B): proportion of sex-bias categories by gene age, chromosome and tissue class (grey=unbiased, blue=male-biased, red=female-biased). (C): Absolute sex bias ($\log_2FC$) by gene age, chromosome and tissue class (blue=significantly male-biased, red=significantly female-biased)

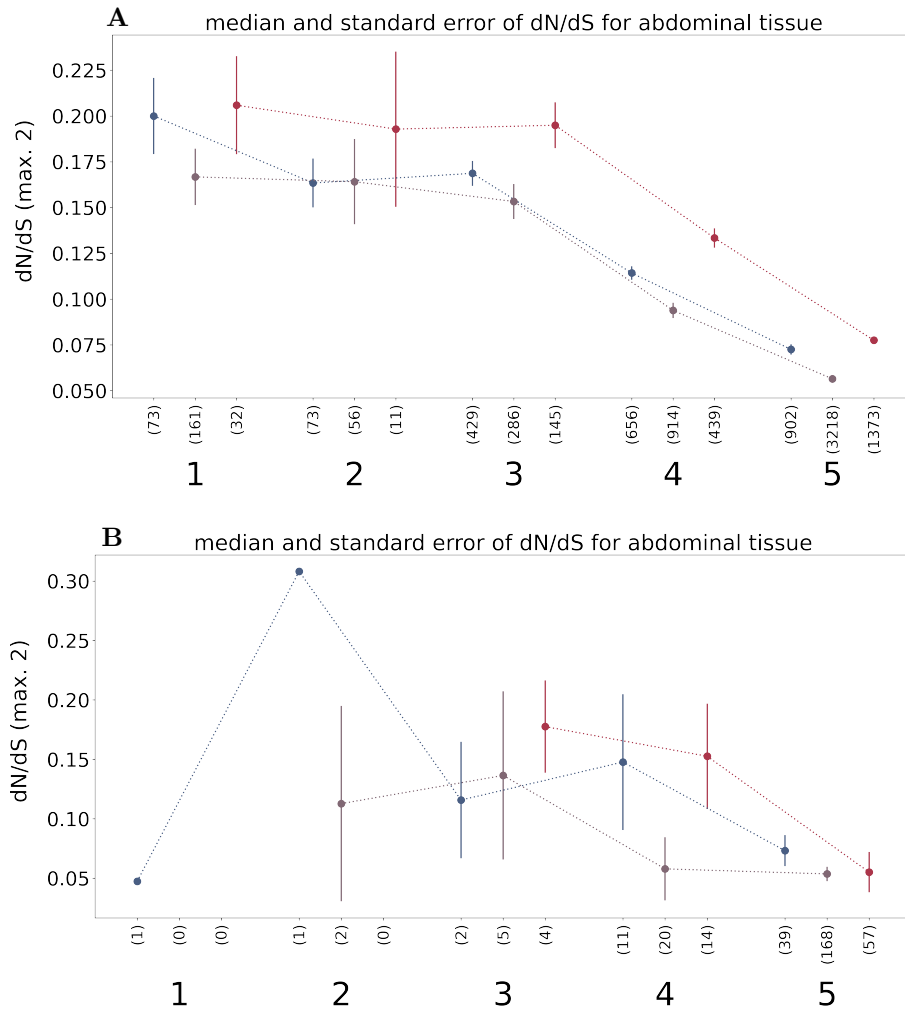


Figure 4: dN/dS by gene age and sex bias: median dN/dS and standard error of autosomal orthologs by gene age and sex-bias category for somatic (abdominal) and reproductive (head + thorax) tissue (grey=unbiased, blue=male-biased, red=female-biased). (A): autosomal orthologs, (B): X-linked orthologs. Gene age ranges from less conserved orthologs (age 1) to highly conserved orthologs (age 5), sample size is shown in parentheses.